

Review of “Invariant measures for the nonlinear stochastic heat equation with no drift term”

Summary

This paper addresses the long time behavior of solutions to the stochastic heat equation (SHE) given by

$$\partial_t u - \frac{1}{2} \Delta u = b(u) \dot{W}, \quad (\text{SHE})$$

where $b = b(x, u)$ is assumed to be bounded in the first argument and globally Lipschitz in the second. The authors identify almost optimal conditions on the spectral measure of the noise via a particular strengthening of Dalang’s condition and expand the admissibility criteria on initial data for guaranteeing the existence of an invariant measure. Their admissibility conditions allow for initial values to be given more generally by measures, thereby extending the classical result by Tessitore and Zabczyk. Moreover, their framework contains the important parabolic Anderson model as a special case.

Although their proof of the existence of invariant measure ultimately relies on the two-step methodology developed by Tessitore and Zabczyk of establishing that the solution laws form Markovian family of transition probabilities with corresponding Feller semigroup, as well as suitable moment bounds that allow for one to carry out a Krylov-Bogoliubov construction of the invariant measure, they invoke an equivalent representation of the solution via the Walsh random field approach. This representation ultimately allows them to obtain more precise conditions for the existence of the invariant measure. A more detailed discussion of their conditions and the types of examples one can consider within their expanded framework are also developed with very interesting calculations.

The paper is written exceptionally well and the arguments are deceptively elementary, especially for obtaining the notoriously difficult issue of obtaining moment bounds, which the reviewer finds to be an indication of the clarity of their analysis and theoretical framework. Ultimately, the authors showcase the power of a Walsh random field approach for studying long time behavior of solutions for space-time stochastically forced PDEs in the context of (SHE), which the reviewer believes may inspire others to consider develop similar approaches in other contexts. The paper is strongly recommended for publication pending the following minor changes.

Comments

1. page 3, paragraph following Definition 1.1: $\mathcal{L}(u(t, \cdot; \mu))$ is missing a parenthesis on the right
2. page 5, the reviewer finds it instructive to view “Dalang’s function” $\Upsilon(\beta)$, for $\beta > 0$, as being an interpolation between its strengthened counterparts, namely:

$$\Upsilon(\beta) \leq 2\Upsilon_\alpha(\beta)^{1/2}\Upsilon(0)^{1/2},$$

which can be proven by simply splitting the integral into near and far fields, then optimizing the constant. This would complement the discussion on page 5/6, as well as the one in Section 3.2

3. page 7, proof of Corollary 1.2, second equation line: there is a * missing on ζ inside the last integral preceding the inequality. The integrand should be $(G(t, \cdot) * \rho)(y)|\zeta^*(y)|^2$
4. page 9, definition of H : the authors should include the definition of H since it is invoked several times in the paper and its properties rely on the restrictions associated to the Dalang-type condition. The reviewer went to reference [3] and does not think it will be difficult to add and will only add clarity to their framework.
5. page 14, the convention in writing integrals is inconsistent. Although this point may seem pedantic, the authors write integrals in at least three different ways, for example as $\int \int dx dy f(x, y)$ or $\int \int f(x, y) dx dy$ or $\int \int dx (f(x, y) dy)$. The author identified that the main convention seems to be to write integrals as $\int \int dx dy f(x, y)$ when calculations or estimates with integrals are being made, while $\int \int f(x, y) dx dy$ when statements are asserted about them such as conditions being stated. The reviewer suggests that the authors make their convention uniform throughout the paper
6. page 26, sentence preceding Section 5.4: “Combing” should be “Combining”